Product Requirements Document (PRD)

AI-Powered Construction Site Intelligence Platform

Version 1.1 | Status: Confirmed Scope — Engineering Handoff Ready

Field	Detail
Document Version	1.1
Project Name	AI-Powered Construction Site Intelligence Platform
Status	Confirmed Scope — Engineering Handoff Ready
Last Updated	July 2025
Changes from v1.0	Material tracking added to delivery logs (material_type, material_quantity, quantity_unit); vendor/driver QR confirmation removed from scope; classroom demo strategy added; out-of-scope clarified between material logging vs. material inventory prediction

1. Executive Summary

The AI-Powered Construction Site Intelligence Platform is a computer-vision-driven safety and operations monitoring system for construction sites. It automates PPE compliance verification at site entry gates, monitors restricted hazardous zones in real time, tracks construction equipment, verifies delivery vehicle counts against manually-reported numbers to detect potential fraud, records what materials arrive on site (type and quantity) per delivery, and centralizes hazard reporting with AI-assisted categorization — all surfaced through a single role-based dashboard for supervisors and administrators.

The platform is built around a shared gate-camera pipeline used by both Module 1A (PPE entry verification) and Module 4A (delivery vehicle counting), reducing hardware cost and operational complexity. It is explicitly positioned as a layered, defense-in-depth system — no single module provides complete coverage, and known limitations are disclosed and mitigated rather than concealed.

This version (v1.1) confirms material inflow tracking as a core feature of Module 4A, adds the classroom/viva demo strategy, and removes vendor/driver QR confirmation from scope with rationale documented.

2. Problem Statement

Construction sites currently rely on manual safety inspections, informal verbal hazard reporting, and paper-based delivery records — all of which are slow, inconsistent, and reactive. Specific problems this platform directly addresses:

Problem	Current State	Impact
PPE compliance checking	Manual visual inspection at gate — one supervisor, hundreds of workers at shift start	Non-compliance goes undetected; queue delays; no audit trail
Restricted zone enforcement	CCTV reviewed after the fact, if at all	Accidents in hazardous zones not prevented in real time
Equipment tracking	Physical walkaround by supervisor	Idle time untracked; location unknown; utilization unreported
Delivery count verification	Supervisor manually counts vehicles and files paper report	Inflated delivery counts go undetected; fraud risk high
Material inflow recording	Paper delivery notes — often lost, incomplete, or forged	No reliable digital record of what materials arrived, when, and how much
Hazard reporting	Verbal or paper forms — no categorization, no audit trail	Hazards unaddressed; no data for safety trend analysis
Site monitoring	Multiple disconnected CCTV systems, no analytics layer	No single source of truth for site safety status

3. Vision & Goals

Vision: Make construction sites measurably safer and more accountable by giving supervisors and administrators real-time, automated visibility into compliance, hazards, deliveries, and site activity — without requiring additional manual inspection staff.

Goal ID	Goal	Module
G1	Reduce PPE non-compliance incidents at site entry	1 / 1A
G2	Provide real-time alerts for restricted zone breaches	2
G3	Track equipment location and idle time automatically	3
G4	Provide tamper-evident, automatic delivery vehicle count verification	4A
G5	Maintain a digital, auditable material inflow log (type + quantity) per delivery	4A
G6	Centralize hazard reporting with AI-assisted categorization	6
G7	Give administrators a single real-time dashboard for all site metrics	7
G8	Build a reusable shared camera pipeline to minimize hardware cost	Cross-cutting

4. Target Users

User Type	Role Description	Primary Interaction
Administrator / Project Manager	Owns the system. Manages users, reviews fraud escalations, oversees all sites, exports compliance and material reports.	Dashboard (all-site), discrepancy escalation queue, user management, zone configuration, reports
Supervisor	Daily site monitoring. Acknowledges violations, logs delivery vehicle counts and material information, manages hazard reports.	Dashboard (site-scoped), violation acknowledgement, delivery + material logging form, hazard report management
Worker	Subject of PPE checks. Submits hazard reports. No dashboard access.	Hazard report submission form only
Vendor / Driver (Passive)	Delivery vehicle detected and counted by gate camera automatically. No app interaction required in current scope.	None — passive actor only

Note: Vendor/Driver QR confirmation was considered and removed from scope. The AI vehicle count (tamper-evident, pre-recorded by gate camera) is the independent verification source. Vendor confirmation would not have been truly independent from a fraud standpoint, as vendors share the same incentive to overstate deliveries.

5. User Personas

Persona	Age / Role	Goals	Pain Points	How Platform Helps
Rajesh Kumar	42, Site Administrator (3 sites)	Catch fraud early, generate compliance reports remotely, track material costs across sites	No real-time visibility; finds out about incidents after the fact; material records are paper-based and unreliable	Dashboard aggregate view; discrepancy escalation queue; exportable material inflow reports
Meena Rao	35, Site Supervisor	Monitor PPE compliance daily, respond to alerts quickly, log deliveries accurately, keep material records	Manual PPE checks cause morning queues; counting delivery vehicles and filing paper material reports is slow and error-prone	Auto-gate PPE check reduces manual inspection; delivery logging form captures count + material in one place; real-time violation alerts

Persona	Age / Role	Goals	Pain Points	How Platform Helps
Arjun Das	27, Construction Worker	Enter site quickly, report hazards without bureaucracy	PPE check queues at shift start; no easy way to report hazards — verbal reports often go unrecorded	Gate check is automatic (no manual ID card); hazard report form is simple image + description + location
Vendor Trucks (Passive)	Daily deliveries	N/A — no app interaction	N/A	Auto-counted at gate; supervisor logs material type/quantity separately

6. User Stories

ID	As a...	I want to...	So that...	Priority
US-1	Worker	Walk through the entry gate and be automatically verified for PPE	I don't need to wait for a manual check or carry a separate ID card	Must Have
US-2	Supervisor	Receive an immediate alert when a worker enters without required PPE	I can intervene before the worker reaches a hazardous area	Must Have
US-3	Supervisor	Receive a real-time alert when someone enters a restricted zone	I can prevent accidents in hazardous areas	Must Have
US-4	Administrator	See the AI-verified vehicle count alongside the supervisor's manually reported count for each delivery	I can detect discrepancies that may indicate inflated delivery reporting	Must Have
US-5	Supervisor	Log the material type (cement/sand/bricks/steel/pipes/other), quantity, and unit for each delivery alongside the vehicle count	There is a complete, searchable digital record of what materials arrived on site	Must Have
US-6	Administrator	View material inflow logs by type and date range	I can track material consumption, verify against purchase orders, and identify discrepancies in material quantities	Must Have
US-7	Worker	Upload a photo and description of a hazard with my location	It gets logged, AI-categorized, and routed to the right person — not lost in verbal communication	Must Have
US-8	Administrator	View all site metrics including PPE compliance, active violations, delivery counts, material inflow, and hazard reports in one dashboard	I don't need to check multiple disconnected systems	Must Have
US-9	Administrator	Review escalated delivery discrepancies without the filing supervisor being involved	Conflict of interest doesn't suppress fraud detection	Must Have
US-10	Supervisor	See camera/system health status on the dashboard	I know when to trust "no alerts" vs when the camera is down	Must Have
US-11	Administrator	Export compliance and material inflow reports by date range	I can share data with project owners and cross-check against procurement records	Should Have
US-12	Supervisor	Filter violations by type, date, and zone	I can quickly find the relevant events during a shift review	Should Have

7. Functional Requirements

Module 1 / 1A — PPE Detection & Entry-Point Verification

ID	Requirement	Priority
FR-1	System shall detect helmet and safety vest presence on a person at the entry gate using YOLOv8 computer vision	Must Have
FR-2	System shall log every gate entry event with: timestamp, compliance status (compliant/violation), violation type (no_helmet/no_vest/both), screenshot URL, and detection confidence score	Must Have
FR-3	System shall send a real-time alert to the supervisor when a non-compliant entry is detected	Must Have
FR-4	System shall use a two-stage detection pipeline: person detection first, then PPE classification on the cropped person region for higher accuracy	Must Have
FR-5	System shall flag "multiple entry — unverified" when more than one person is detected in frame during a single gate cycle (tailgating prevention)	Should Have
FR-6	System shall optionally integrate with turnstile/gate hardware to block non-compliant entry — configurable per site	Nice to Have

Module 2 — Restricted Area / Zone Monitoring

ID	Requirement	Priority
FR-7	System shall allow administrators to define restricted zones as polygon coordinates per site	Must Have
FR-8	System shall detect person presence inside any defined restricted polygon zone in real time using the pretrained YOLOv8 person class (no fine-tuning required)	Must Have
FR-9	System shall generate an immediate alert and log a violation record when a person enters a restricted zone	Must Have
FR-10	System shall support multiple zone types: crane_operating, high_voltage, excavation, chemical_storage	Must Have
FR-11	System shall support zone-specific PPE rules (e.g., harness required in addition to helmet+vest for crane zones) — configurable by admin	Should Have

Module 3 — Equipment Monitoring

ID	Requirement	Priority
FR-12	System shall detect and classify construction equipment type (excavator, bulldozer, crane, cement mixer, truck) via YOLOv8	Must Have
FR-13	System shall log equipment last-seen location and timestamp from camera detection	Must Have
FR-14	System shall track consecutive idle time (minutes without movement detected) per equipment unit	Must Have
FR-15	System shall display equipment status (active/idle/offline) on the dashboard	Must Have

Module 4A — Delivery Vehicle Count Verification + Material Inflow Logging

ID	Requirement	Priority
FR-16	System shall automatically count delivery vehicles entering through the gate camera and record the count with a timestamp — independent of any supervisor action	Must Have
FR-17	The AI vehicle count must be written to the database before supervisor manual entry is possible, establishing it as the independent, tamper-evident ground truth	Must Have
FR-18	System shall provide a delivery logging form for supervisors to enter: manual vehicle count, material type, material quantity, and quantity unit	Must Have
FR-19	Material type field shall be constrained to an allowed list: cement, sand, bricks, steel, pipes, other — free-text entry not permitted	Must Have
FR-20	Quantity unit field shall be constrained to: bags, tons, cubic_meters, pieces	Must Have

ID	Requirement	Priority
FR-21	System shall compare AI vehicle count against manual reported count and track the mismatch count per supervisor per gate	Must Have
FR-22	System shall escalate a discrepancy alert to the Administrator (not the filing Supervisor) only when repeated/patterned mismatches are detected — a single mismatch shall NOT trigger escalation	Must Have
FR-23	discrepancy_flag shall be based on vehicle count mismatch only — material type and quantity are informational and shall NOT affect fraud detection logic	Must Have
FR-24	System shall optionally classify vehicle type (mixer/tipper/open_bed/flatbed) as a soft cross-check against logged material type — flagging unusual combinations (e.g., cement logged but open-bed truck detected)	Should Have
FR-25	System shall provide a material inflow log view filterable by date range, gate, and material type — showing totals per material type for the selected period	Must Have
FR-26	If supervisor submits manual count before AI count is logged, system shall set status to "Pending AI Count" and hold discrepancy comparison — shall not compare against null	Must Have

Module 6 — Hazard Reporting

ID	Requirement	Priority
FR-27	System shall allow workers to submit hazard reports via a form with: image upload (optional), description (required), location (required)	Must Have
FR-28	System shall AI-categorize submitted hazard reports into: fire_risk, unsafe_wiring, water_leakage, scaffold_damage, structural_cracks, uncategorized	Must Have
FR-29	Reports with AI categorization confidence below threshold shall be placed in an "uncategorized" queue for supervisor manual review	Must Have
FR-30	System shall allow supervisors to acknowledge or escalate hazard reports	Must Have
FR-31	Workers shall only be able to submit reports — they shall not be able to view other workers' reports or any other system data	Must Have

Module 7 — AI Dashboard

ID	Requirement	Priority
FR-32	Dashboard shall display the following KPI cards: PPE Compliance Rate (%), Live Worker Count, Active Violations, Unacknowledged Violations, Today's AI Vehicle Count, Open Hazard Reports, Escalated Discrepancies	Must Have
FR-33	Dashboard shall display a live alert feed showing recent violations and zone breaches in real time	Must Have
FR-34	Dashboard shall display camera/system health status per gate and zone — "no alerts" must never be ambiguous with "camera is offline"	Must Have
FR-35	Dashboard shall display a today's material inflow summary (bar chart: material type vs. quantity received today)	Must Have
FR-36	Admin dashboard shall display an escalation queue banner prominently when unresolved discrepancies exist	Must Have
FR-37	All dashboard data shall be scoped by role: Supervisors see own site only; Admins see aggregate across all sites	Must Have

Cross-Cutting Requirements

ID	Requirement	Priority
FR-38	System shall enforce role-based access control (RBAC) at the API layer for all endpoints — Admin, Supervisor, Worker roles	Must Have
FR-39	System shall support multiple sites with site-scoped data isolation for supervisors	Must Have
FR-40	All violation and delivery log records shall be append-only — no update or delete operations permitted on audit tables	Must Have

8. Non-Functional Requirements

ID	Requirement	Target / Notes
NFR-1	Detection-to-alert latency	< 5 seconds from camera detection to dashboard alert appearing
NFR-2	API response time	< 300ms for standard CRUD endpoints; < 2s for dashboard summary aggregation
NFR-3	System availability	99% uptime during site working hours
NFR-4	Authentication	JWT with 15-minute access tokens and 7-day rotating refresh tokens
NFR-5	Authorization	Role checks enforced server-side on every API endpoint — never trust client-side role claims
NFR-6	Scalability	Stateless API layer horizontally scalable; AI inference workers scale independently
NFR-7	Audit integrity	Append-only audit tables with no update/delete endpoints — tamper-evident by design
NFR-8	Data security	HTTPS/TLS everywhere; secrets in environment variables; no plaintext passwords
NFR-9	Multi-site support	Platform supports multiple sites with data isolation per site
NFR-10	Demo/offline capability	System must be fully runnable via Docker Compose locally without cloud dependency — required for classroom demo

9. Business Rules

Rule ID	Business Rule	Rationale
BR-1	Discrepancy escalation alerts from Module 4A are sent to the Administrator role only — the filing Supervisor is never notified of their own discrepancy escalation	Conflict of interest — the supervisor filing the delivery count is the party who may be inflating it
BR-2	A single vehicle count mismatch does not trigger escalation — only repeated/patterned mismatches (at the same gate by the same supervisor) trigger escalation	Avoids false accusations from ordinary miscounts; protects supervisor credibility on genuine errors
BR-3	discrepancy_flag is computed from vehicle count mismatch only — material type and quantity fields are informational and do not affect fraud detection logic	AI cannot reliably verify cargo contents through a closed truck body — only count is AI-verified
BR-4	Material type and quantity are entered manually by the supervisor — the AI layer verifies vehicle count (and optionally vehicle type) only	Reliable cargo content identification from external camera view is outside the scope of this system's CV capability
BR-5	AI vehicle count is written to the database before supervisor manual entry is accepted — it serves as the independent ground truth	Prevents supervisor from knowing the AI count before filing their own number
BR-6	Workers cannot self-register — all user accounts are created by the Administrator	Prevents unauthorized access; ensures site affiliation is verified before account creation
BR-7	Historical audit records (violations, delivery logs) are preserved even if the associated user or gate is deactivated (ON DELETE SET NULL, not CASCADE)	Evidence must never be deleted by removing a related entity
BR-8	Gate PPE check failures may optionally trigger physical turnstile lock — configurable per site, not enforced by default	Not all sites have turnstile hardware; system should work as an alerting layer even without hardware integration

10. Success Metrics (KPIs)

KPI	Measurement Method	Target
PPE detection model accuracy (mAP)	Evaluated on held-out test set after training	>= 85%
False positive rate on gate PPE checks	Manual audit of flagged vs. actual violations sample	< 10%
Detection-to-alert latency	Timestamp: detection event vs. dashboard toast appearance	< 5 seconds
Material inflow log completeness	% of delivery events where material_type is logged	>= 95%
Discrepancy detection coverage	Confirmed fraud cases caught vs. total confirmed fraud cases in audit	Measurable improvement over manual-only baseline
Dashboard uptime	Monitoring health-check endpoint	99% during working hours
Hazard report categorization accuracy	Spot-check AI categories vs. supervisor manual review sample	>= 80% correct category assignment
System camera health visibility	Supervisors report "no ambiguity between no alerts vs. camera offline"	Zero reports of missed alerts due to undetected camera downtime

11. Assumptions

ID	Assumption	Impact if Wrong
A1	A single gate camera can serve both Module 1A (PPE) and Module 4A (vehicle counting) simultaneously or in close sequence	Would require separate cameras per module — hardware cost increase
A2	Supervisors will consistently log material type and quantity for each delivery — system cannot force this if supervisor skips the step	Material inflow log completeness KPI will be below 95%; incomplete data reduces site-manager value
A3	Cloud deployment target is flexible (AWS / Azure / Render) — Docker containers make this vendor-agnostic	Minimal — any major cloud supports Docker
A4	No native mobile app in v1 — web dashboard on tablet/desktop is sufficient for operational use	Workers without desktop access cannot submit hazard reports on mobile — need a mobile-responsive form at minimum
A5	The classroom/viva demo will use pre-recorded video clips, not a live construction site camera	Pre-recorded video is standard academic practice and fully accepted for demo purposes
A6	Material inflow logging is a manual data-entry task, not AI-detected — the AI only verifies vehicle count and optionally vehicle type	If stakeholders expect AI to detect cargo contents, this must be explicitly corrected — it is a disclosed limitation, not a gap

12. Constraints

Constraint	Details
Out of scope: Material Inventory Prediction	Forecasting future material shortages from historical inflow data is NOT in v1 scope. Material inflow logging (recording what arrived) IS in scope. These are distinct features — logging is data capture; prediction is an ML forecasting module.
Out of scope: Project Progress Estimation	Estimating construction completion % from periodic images is NOT in scope. This requires specialized image segmentation and labeled historical images not available for a prototype.
Out of scope: Drone surveillance	Not in v1. Listed as future enhancement.
Out of scope: IoT sensor integration	Not in v1. Listed as future enhancement.
Out of scope: Vendor/Driver QR confirmation	Removed from scope. Vendor confirmation is not truly independent from a fraud-detection standpoint — vendors share the same incentive to overstate

Constraint	Details
	deliveries as a colluding supervisor. The AI vehicle count is the independent source.
Out of scope: Mobile app	v1 is web-only. Hazard report form must be mobile-responsive for worker access from phones.
AI limitation: cargo content verification	The AI pipeline cannot reliably identify what is inside a closed truck body from an external camera. Material type and quantity are always manually entered by the supervisor.

13. Risks & Mitigations

Risk ID	Risk	Likelihood	Impact	Mitigation Strategy
R-1	Low-light or adverse weather degrades gate camera PPE detection accuracy	Medium	High	IR/night-vision cameras; covered gate canopy; image preprocessing (histogram equalization, denoising) before inference
R-2	Tailgating at gate — multiple workers entering simultaneously	High	Medium	Physical single-file channeling (narrow turnstile/lane); person-count check flags multi-entry events for supervisor manual verification
R-3	Workers remove PPE after passing the entry gate check	High	Medium	Entry gate check is first line of defense only; periodic spot-check cameras inside key work zones provide secondary coverage. Disclosed as known limitation.
R-4	False fraud accusations from individual delivery count mismatches	Low	High (trust damage)	Pattern-based escalation only (not single-event); threshold configurable by admin; clear UI explanation that one mismatch does not flag
R-5	Supervisors skip material type/quantity fields when logging deliveries	Medium	Medium	Make material fields required in the delivery logging form; track completeness as a KPI; supervisor training
R-6	AI model overfits to small training dataset — poor real-world generalization	Medium	High	Document dataset size and source explicitly; report evaluation metrics honestly; train on site-specific PPE colors/styles where possible
R-7	Camera goes offline silently — supervisors interpret "no alerts" as "all clear"	Medium	High	Camera health status shown persistently on dashboard; separate alert for camera/AI pipeline downtime; time window marked as "unmonitored" in audit log
R-8	Demo fails due to venue Wi-Fi or cloud dependency	Medium	High (academic grade)	Full local Docker Compose setup — demo runs entirely offline; pre-recorded clips instead of live camera feed
R-9	Privacy/legal concerns about continuous worker surveillance	Low-Medium	Medium	Explicit worker consent at onboarding; data retention policy defined; avoid biometric identification in v1 (use RFID/badge if identity needed)
R-10	Examiner questions AI capability overclaiming	Medium	High (grade impact)	All limitations explicitly documented in report Limitations section; evaluation metrics shown honestly; "why AI" justification prepared per module

14. Future Scope

Feature	Description	Dependency
Material Inventory Prediction	Forecast material consumption and predict reorder dates using historical inflow data logged by Module 4A — this is the natural ML extension of the logging feature already built	Requires 3–6 months of logged delivery data as training input

Feature	Description	Dependency
Project Progress Estimation	Periodic image capture + comparison to estimate construction completion percentage by stage (foundation, structural, roofing)	Requires labeled historical images per construction stage — not publicly available at scale
Drone-Based Surveillance	Extend zone monitoring and PPE checking beyond fixed-camera coverage using drone feeds	Drone hardware + flight path automation
IoT Sensor Integration	Supplement CV-based detection with environmental sensors (temperature, CO2, noise levels) in hazardous zones	IoT device procurement + integration layer
Mobile Application	Dedicated mobile app for hazard reporting (workers) and alert monitoring (supervisors) — v1 web app is mobile-responsive but not a native app	React Native or Flutter build
Vendor/Driver QR Confirmation	Optional identity-only layer — vendor scans QR on arrival to confirm which vendor/driver is present (not for fraud detection — the AI count serves that purpose)	Vendor onboarding + QR scanning at gate
Multi-Site Management Console	Unified admin console to manage and compare metrics across many construction sites for large enterprise clients	Multi-tenant architecture refactor
AI Accident Risk Prediction	Use historical violation and hazard data to predict high-risk times/locations on site	Sufficient historical data volume required
BIM Integration	Overlay safety events onto a Building Information Model for spatial analysis	BIM software API access

15. Classroom / Viva Demo Strategy

Since a live construction site is not available for the academic demo, the following approach is confirmed as the standard, accepted method for academic project demonstrations:

Demo Step	Action	What It Shows
Step 1	Show empty dashboard (idle state)	System is live; KPI cards at zero; camera health shown as online
Step 2	Play Clip 1: worker without helmet entering gate	PPE violation logged in real time; alert appears on dashboard; screenshot captured
Step 3	Play Clip 2: worker with full PPE entering gate	Compliant entry logged; no alert; compliance rate increments
Step 4	Play Clip 3: vehicle entering gate	AI vehicle count increments automatically with timestamp
Step 5	Supervisor logs delivery: manual count (intentionally mismatched) + material type "cement" + quantity 500 bags	Delivery form with material fields; discrepancy comparison triggered; pattern logic shown
Step 6	Submit worker hazard report (image + description + location)	AI category assigned; report appears in dashboard with category badge
Step 7	Switch to Admin view — show discrepancy escalation screen	Admin-only routing; side-by-side AI vs. manual count; material info shown; supervisor name visible
Step 8	Play deliberately low-light or occluded clip	Proactively show a failure/limitation; explain confidence threshold and manual review queue
Step 9	Show evaluation metrics slide (precision/recall/mAP)	Honest performance disclosure — your own numbers from your own test set

All clips are pre-recorded video files fed through the YOLOv8 inference pipeline — the pipeline does not distinguish between a live RTSP stream and a video file. The entire demo runs locally via Docker Compose with no dependency on venue Wi-Fi or cloud connectivity.